CANALRUNG · DESIGN DECISION RECORD

CanalRung TRL 2 review decisions

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	CNR-DDR-001	Version	0.1
Status	Draft	Date	2026-10-03
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/canalrung		
Source	docs/decisions/0001-trl2-review-decisions.md @ ce4cb4e		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-03	Amish Chadha	TRL 2 review items D1 to D14, decided as recommended under Amish's 2026-10-03 pre-approval

0001: TRL 2 review decisions

- **Date:** 2026-10-03
- **Status:** accepted. Decided by Amish under his pre-approval of 2026-10-03: "start with the first 14 repos from the list of 29 projects. I pre-approve the batch runs along with any recommendations you come up with. I also accept any cost overruns or variations from the assumed scope cost."

Context

The scaffold (CNR-PRB-001, CNR-PRC-001 and CNR-REQ-001, all v0.1) left five open questions and several choices undecided: bank heights and flows, sink or float, the load standard, theft, the anchor type, the rung and rope construction, the throw line and the partners. Populating the concept to TRL 2 needed an answer to each. Because Amish pre-approved every recommendation for this batch, each item below is decided as recommended rather than left "Proposed, awaiting Amish". Choices that touch safety take the conservative option, and the record says what evidence would relax them. Partners and regions are named as the first candidates to approach, not as agreed.

Options considered

Table 1. Options for each item.

#	Item	Options
D1	Design envelope	(a) 1.5:1 slopes to 3.0 m (needs a 5.4 m ladder and about 6 kg); (b) 1.5:1 slopes to 2.0 m and vertical walls to 3.0 m with a 4.0 m ladder; (c) vertical walls only
D2	Anchor	(a) ground stake driven at the time; (b) post loop around any nearby post; (c) cross bar across the bank; (d) the station post itself, set in concrete, with the ladder linked to it in advance
D3	Rung	(a) 25.4 x 1.65 mm aluminium tube (lighter); (b) 28.6 x 1.65 mm aluminium tube (stronger); (c) solid HDPE rod (floats but about 6 kg for the set); (d) GFRP tube
D4	Rope and joint	(a) 10.5 mm EN 1891 type A kernmantle through the rung ends, overhand stopper knot under and seizing over each end; (b) polypropylene rope with clamps; (c) laid rope with rungs seized between the strands (Jacob's ladder)
D5	Wall stand-offs	(a) on every rung (about 1.3 kg); (b) on every third rung (0.42 kg); (c) none
D6	Bottom end	(a) weighted to sink; (b) floating; (c) a soft weight on a short tail with floating rungs above it
D7	Throw line	(a) fitted; (b) left out of the first prototype
D8	Storage	(a) a bought UV-stabilised 70 L box on the post, no lock; (b) a made steel cabinet; (c) a locked box
D9	Load standard	(a) certify to a ladder or life-saving standard now; (b) rope to EN 1891 type A, rung and anchor targets as in CNR-REQ-001, proof-loaded at TRL 4, no certification claimed
D10	R2 mass target	(a) meet 4.0 kg with lighter rungs and fewer stand-offs; (b) keep the safer rung and report R2 as not met
D11	Theft	(a) lock; (b) no lock, tamper tag and inspection; (c) no rope or aluminium (not possible)
D12	Shared blocks	CalRig for proof loads; LaneSkiff boarding step
D13	Co-design partner	Western US irrigation districts; Punjab canal authorities; water-safety councils
D14	Pitch, problem and value-engineering target	Change or keep

Decision

Table 2. Items decided, 2026-10-03.

#	Decision	Why	What would relax it
D1	(b): water to 2.0 m below the coping on 1.5:1 slopes and to 3.0 m on vertical walls; currents to 2.0 m/s	Keeps the ladder near the 4 kg throwing target; covers the vertical-walled and steep reaches where self-rescue is hardest. A 6 m variant for deeper sloped canals is a later option	Partner district data on its reaches
D2	(d): the post is the anchor; forged M16 eye nut on an M16 bolt through the post, 250 mm above ground, ladder linked in advance	A bystander cannot set a 4.5 kN stake in 30 s; a hand-held top pulls people in. The conservative option	A tested portable anchor for a vehicle kit
D3	(b): 28.6 x 1.65 mm 6061-T6, 360 mm long	0.56 of yield at 1.5 kN against 0.73 for the lighter tube. Safety before mass	Proof-load results showing the lighter tube keeps a two-thirds margin
D4	(a): EN 1891 type A 10.5 mm, polyester preferred; nylon bushes in the rung holes; overhand knot under, seizing over	Rated rope with known knotted strength; a joint anyone can tie and inspect; polypropylene is weak and degrades in sun	
D5	(b): rungs 3, 6, 9 and 12, 95 mm from rung axis to wall	Hand room on a flat wall at a third of the mass of (a)	Tank trials showing climbers manage with fewer
D6	(c): soft 0.5 kg steel shot pouch, 280 mm below the bottom rung, rungs floating	The end goes down and stays put; the rungs stay at the surface; nothing hard is thrown near the person	Throw trials
D7	(b): no throw line on the first prototype	A second line can tangle and is an entanglement hazard in moving water. Conservative	Throw trials showing a line is needed to steer the ladder
D8	(a): bought bright 70 L UV-stabilised box clamped to the post, drain holes, rope notch, blunt-tip knife under the lid	Cheap, replaceable, visible; keeps the rope dark	
D9	(b)	Certification is beyond TRL 3; the targets give margins over the design climber and will be proof-loaded with CalRig	
D10	(b): keep the D3 rung; R2 reported as not met (4.48 kg against 4.0 kg)	The rung margin protects the climber; 0.5 kg more may not matter to a thrower. Savings to try are in the register	Throw trials with the 4.5 kg ladder
D11	(b): no lock; numbered tamper tag on the latch; the partner district inspects on a set schedule and after each use	A locked rescue box fails when it is needed	Theft data from a pilot
D12	CalRig named as the proof-load rig for TRL 4 (candidate shared block); the LaneSkiff boarding step stays outside this repository	Keeps this repository to one product	
D13	First candidates to approach: the Kittitas Reclamation District (Washington) as co-design partner and the Water Safety Council of Fresno	Both publish canal safety guidance cited in CNR-PRB-001	

#	Decision	Why	What would relax it
	County as water-safety partner; neither contacted		
D14	No change: pitch and problem as scaffolded; `budget_usd` stays 500 as the value-engineering target	Cost overruns are accepted under the pre-approval and reported, not used to change the target	

Consequences

- CNR-PRB-001, CNR-PRC-001 and CNR-REQ-001 are revised. R1 is restated for the D1 envelope, R6 for the fixed station anchor of D2, and R10 is added for D6 and D8.
- The design is modelled in `cad/src/model.py` on these decisions and then made constructable in CNR-DDR-002.
- `project.yaml` : only the TRL fields, `design_state` and the evidence list change; `budget_usd` , the pitch and the problem are unchanged.
- Nothing in this record authorizes building or testing. TRL 4 is outside the current portfolio phase.

> **Safety:** Every choice above that touches safety (D2, D3, D6, D7, D11) took the more conservative option. None of them makes the ladder safe without proof loads and supervised trials.